

# Qatar-9b

R-2295

Benchmark run · workflow W8-benchmark-deep · record deep-bench\_Qatar-9\_251222 · created 2026-08-17T01:14:24+00:00

BENCHMARK: ACCURATE — fit consistent with published values

## Results

|                                                   |                                 |
|---------------------------------------------------|---------------------------------|
| Mid-Transit Time (Tmid)                           | 2461031.9703 +/- 0.0059 BJD_TDB |
| Ratio of Planet to Stellar Radius (Rp/R*)         | 0.151 +/- 0.021                 |
| Transit depth (Rp/Rs)^2                           | 2.29 +/- 0.65 [%]               |
| Orbital Inclination (inc)                         | 89.95 +/- 1.31                  |
| Airmass coefficient 1 (a1)                        | 0.867 +/- 0.023                 |
| Airmass coefficient 2 (a2)                        | 0.12 +/- 0.11                   |
| Scatter in the residuals of the lightcurve fit is | 2.7 %                           |
| Best Comparison Star                              | #3 - [402, 267]                 |
| Optimal Method                                    | PSF photometry                  |
| Transit Duration (day)                            | 0.0775 +/- 0.0019               |

## Accuracy vs published ephemeris (ExoClock/NEA)

|                           |                                          |
|---------------------------|------------------------------------------|
| Rp/R* fit vs published    | 0.151 ± 0.021 vs 0.1489 (deviation 0.1σ) |
| Duration fit vs published | 0.0775 d vs 0.0779 d (deviation 0.22σ)   |
| Mid-time O–C              | 3.5 min                                  |
| Depth SNR                 | 7.1                                      |
| Residual scatter          | 2.7 %                                    |

## Light curve

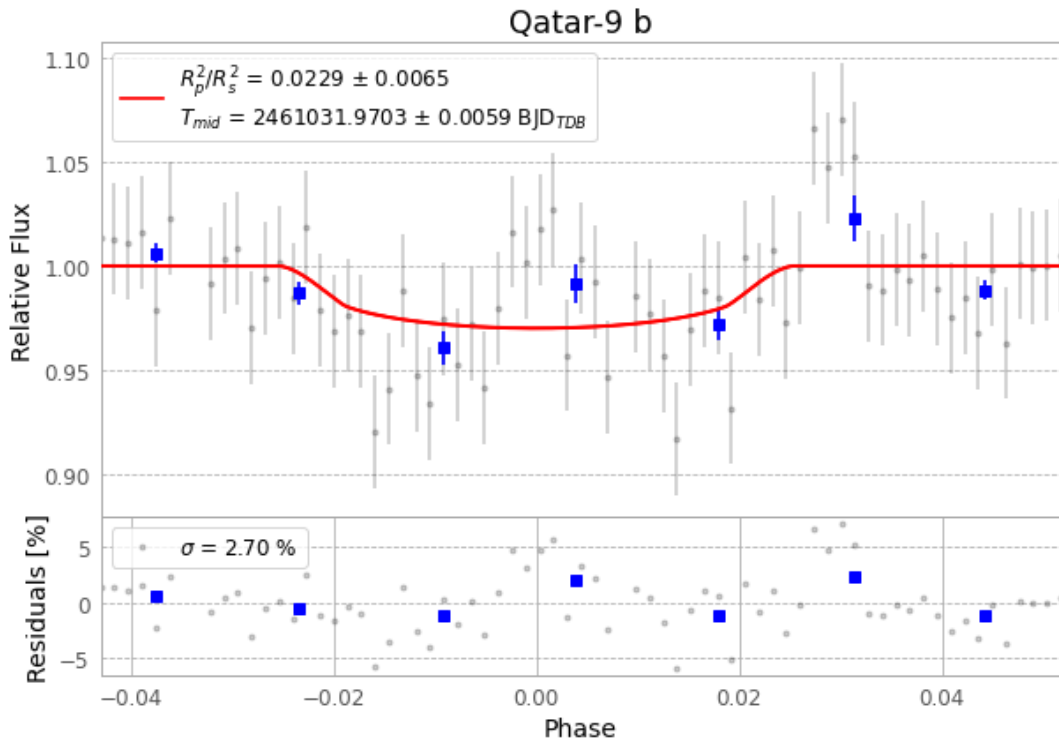


Figure 1 — FinalLightCurve\_Qatar-9 b\_2025-12-22.png (SHA-256 83d9d11eb4e4e2ad...)

## Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [447, 299], 3 comparison star(s). Exact configuration: parameters.inits\_json in record.json (SHA-256 2ecfbc37f06d91dd...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline\_commit 1dc3d22

## Data provenance

71 FITS frames (47 MB), Qatar-9251222093615.FITS → Qatar-9251222130615.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-9-251222.log (45261f806f63...), FinalLightCurve\_Qatar-9 b\_2025-12-22.png (83d9d11eb4e4...), FinalLightCurve\_Qatar-9 b\_2025-12-22.csv (2d11cdd110ea...)

## Compute resources

Wall time 107.9 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-17 01:14Z.